

```
*****Default*****
hostapd
pico-sdk
picotool
Testing pico_w-mqtt
pi4-3 pi4-37 & pi4-20
07/07/25
*****Default*****
```

See pico_w-mqtt/doc/pi4-20.txt & pico_w-mqtt/doc/pi4-20
You need several environment variables

```
export WIFI_SSID="WIFI-SSID"
export WIFI_PASSWORD="WIFI-PASSWORD"
```

```
export PIPELINEC_REPO="$HOME/PipelineC"
export PICO_EXTRAS_PATH=/home/devel/pico-extras
```

```
export PICO_SDK_PATH=/home/devel/pico2/pico-sdk
#export PICO_2350=1
```

```
export FREERTOS_KERNEL_PATH=/home/devel/pico2/freertos-lib/FreeRTOS-Kernel
```

```
export PICTOOL_FETCH_FROM_GITHUB=/home/devel/pico2/BUILT_PICOTOOL
```

```
devel@wifiextender:~ $ ifconfig
```

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.42.1 netmask 255.255.255.0 broadcast 192.168.42.255
    inet6 fe80::c831:9f7f:f5b9:4a5d prefixlen 64 scopeid 0x20<link>
    ether 00:0e:c6:8f:a2:6c txqueuelen 1000 (Ethernet)
    RX packets 88715 bytes 10862968 (10.3 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 187830 bytes 261245596 (249.1 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.235 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 2600:1700:69f0:42c0:5321:921c:c79a:7c64 prefixlen 64 scopeid 0x0<global>
    inet6 2600:1700:69f0:42c0::40 prefixlen 128 scopeid 0x0<global>
    inet6 fe80::f1ab:6da4:88d8:e3ed prefixlen 64 scopeid 0x20<link>
    ether b8:27:eb:2b:eb:c2 txqueuelen 1000 (Ethernet)
    RX packets 203163 bytes 261293049 (249.1 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 89075 bytes 12131497 (11.5 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
Contents of devel@pi5-90:~ $ ./scripts-rpi/route/2nd_level.sh
```

```
#!/bin/bash
```

```
route
```

```
sudo route add -net 192.168.42.0 netmask 255.255.255.0 gw 192.168.1.235 eth0
```

```
route
```

```
devel@pi5-90:~ $ ./scripts-rpi/route/2nd_level.sh
```

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
default	dsldevice.attlo	0.0.0.0	UG	100	0	0	eth0
192.168.1.0	0.0.0.0	255.255.255.0	U	100	0	0	eth0

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use	Iface
default	dsldevice.attlo	0.0.0.0	UG	100	0	0	eth0
192.168.1.0	0.0.0.0	255.255.255.0	U	100	0	0	eth0
192.168.42.0	wifiextender	255.255.255.0	UG	0	0	0	eth0

```
devel@pi5-90:~ $ ./scripts-rpi/route/2nd_level.sh
```

```
devel@pi4-37:~ $ git clone https://github.com/develone/pico_w-mqtt.git -b dev-test-RP-FK
commit 7b82c43640a328a62888483db5cbb112343520d (HEAD -> dev-test-RP-FK, origin/dev-test-RP-FK)
```

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Date: Thu Jul 3 11:52:56 2025 -0600

need infor

picotool is needed for builds and was installed on pi4-3, pi4-37, and pi4_20.

```
devel@pi4-37:~/pico2/BUILT_PICOTOOL/picotool $ cd build/
```

```
devel@pi4-37:~/pico2/BUILT_PICOTOOL/picotool/build $ cmake ..
```

-- The C compiler identification is GNU 10.2.1

-- The CXX compiler identification is GNU 10.2.1

-- Detecting C compiler ABI info

-- Detecting C compiler ABI info - done

-- Check for working C compiler: /usr/bin/cc - skipped

-- Detecting C compile features

-- Detecting C compile features - done

-- Detecting CXX compiler ABI info

-- Detecting CXX compiler ABI info - done

-- Check for working CXX compiler: /usr/bin/c++ - skipped

-- Detecting CXX compile features

-- Detecting CXX compile features - done

Using PICO_SDK_PATH from environment ('/home/devel/pico2/pico-sdk')

./usr/local/share/picotool/

-- Using the single-header code from

/home/devel/pico2/BUILT_PICOTOOL/picotool/lib/nlohmann_json/single_include/

-- Found Python3: /usr/bin/python3.9 (found version "3.9.2") found components: Interpreter

-- Performing Test C_COMPILER_SUPPORTS_WFORMAT_SIGNEDNESS

-- Performing Test C_COMPILER_SUPPORTS_WFORMAT_SIGNEDNESS - Success

-- Found PkgConfig: /usr/bin/pkg-config (found version "0.29.2")

-- Checking for module 'libusb-1.0'

-- Found libusb-1.0, version 1.0.24

-- Found LIBUSB: /usr/lib/aarch64-linux-gnu/libusb-1.0.so

-- Configuring done

-- Generating done

-- Build files have been written to: /home/devel/pico2/BUILT_PICOTOOL/picotool/build

```
devel@pi4-37:~/pico2/BUILT_PICOTOOL/picotool/build $ make
```

```
devel@pi4-37:~/pico2/BUILT_PICOTOOL/picotool/build $ sudo make install
```

```
[ 54%] Built target mbedcrypto
[ 56%] Built target errors
[ 57%] Built target elf
[ 64%] Built target mbedx509
[ 72%] Built target mbedtls
[ 74%] Built target bintool
[ 75%] Performing build step for 'otp_header_parser'
[100%] Built target otp_header_parse
[ 76%] No install step for 'otp_header_parser'
[ 76%] Completed 'otp_header_parser'
[ 79%] Built target otp_header_parser
[ 80%] Built target generate_otp_header
[ 81%] Performing build step for 'xip_ram_perms'
[ 81%] No install step for 'xip_ram_perms'
[ 82%] Completed 'xip_ram_perms'
[ 86%] Built target xip_ram_perms
[ 87%] Performing build step for 'flash_id'
[ 88%] No install step for 'flash_id'
[ 88%] Completed 'flash_id'
[ 91%] Built target flash_id
[ 93%] Built target binary_data
[ 94%] Built target elf2uf2
[100%] Built target picotool
Install the project...
-- Install configuration: "Release"
-- Installing: /usr/local/bin/picotool
-- Installing: /usr/local/lib/cmake/picotool/picotoolTargets.cmake
-- Installing: /usr/local/lib/cmake/picotool/picotoolTargets-release.cmake
-- Installing: /usr/local/lib/cmake/picotool/picotoolConfig.cmake
-- Installing: /usr/local/lib/cmake/picotool/picotoolConfigVersion.cmake
-- Installing: /usr/local/share/picotool/rp2350_otp_contents.json
-- Installing: /usr/local/share/picotool/xip_ram_perms.elf
devel@pi4-37:~ $ cd pico_w-mqtt
There are three scripts 1remote.sh, 3remotes.sh and 6remotes.sh
```

```
#!/bin/bash
```

```
./build_cli.sh
```

```
rm -rf remote5
```

```
mkdir remote5
```

```
cd remote5
```

```
cp ../exe-ocd.sh .
```

```
cmake -DPICO_BOARD=pico_w -DHOSTNAME="remote5" -DWIFI_SSID="WIFI-SSID" -  
DWIFI_PASSWORD="WIFI_PASSWORD" ..
```

```
make
```

```
cd ../
```

```
echo "Done"
```

```
root@pi4-37:/home/devel# less /etc/hostapd/hostapd.conf
```

```
interface=wlan0
```

```
driver=nl80211
```

```
ssid=WIFI-SSID
```

```
hw_mode=g
```

```
channel=7
```

```
wmm_enabled=0
```

```
macaddr_acl=0
```

```
auth_algs=1
```

```
ignore_broadcast_ssid=0
```

```
wpa=2
```

```
wpa_passphrase=WIFI-PASSWORD
```

```
wpa_key_mgmt=WPA-PSK
```

```
wpa_pairwise=TKIP
```

```
rsn_pairwise=CCMP
```

```
devel@pi4-37:~/pico_w-mqtt/pi_tcp_tests $ ifconfig eth0
```

```
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.42.105 netmask 255.255.255.0 broadcast 192.168.42.255
    inet6 fe80::f749:56c8:dcef:6b0d prefixlen 64 scopeid 0x20<link>
    ether b8:27:eb:07:c3:34 txqueuelen 1000 (Ethernet)
    RX packets 1168 bytes 120195 (117.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1060 bytes 176184 (172.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
devel@pi4-37:~/pico_w-mqtt/pi_tcp_tests $ ifconfig wlan0
```

```
wlan0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.32.1 netmask 255.255.255.0 broadcast 192.168.32.255
    inet6 fe80::b321:ad1f:1ecb:d9b2 prefixlen 64 scopeid 0x20<link>
    ether b8:27:eb:52:96:61 txqueuelen 1000 (Ethernet)
    RX packets 12273 bytes 504328 (492.5 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 25172 bytes 37424267 (35.6 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

```
devel@pi4-37:~ $ ./scripts-rpi/mosquitto/mos.sh
```

```
1751806446: mosquitto version 2.0.11 starting
```

```
1751806446: Config loaded from /etc/mosquitto/mosquitto.conf.
```

```
1751806446: Opening ipv4 listen socket on port 1883.
```

```
1751806446: Opening ipv6 listen socket on port 1883.
```

```
1751806446: mosquitto version 2.0.11 running
```

```
devel@pi4-37:~ $ 1751806447: New connection from 192.168.32.106:57277 on port 1883.
```

```
1751806447: New client connected from 192.168.32.106:57277 as remote5 (p2, c1, k10, u'testuser').
```

```
1751806477: New connection from ::1:60742 on port 1883.
```

```
1751806477: New client connected from ::1:60742 as auto-93647E86-6EDF-1B7C-8D85-315DFE4CB7E0 (p2, c1, k60, u'testuser').
```

```
1751806879: New connection from 192.168.42.105:45632 on port 1883.
1751806879: New client connected from 192.168.42.105:45632 as auto-7944AF4C-6C68-BC61-
DF2F-3A203CFC51FD (p2, c1, k60, u'testuser').
1751806879: Client auto-7944AF4C-6C68-BC61-DF2F-3A203CFC51FD disconnected.
1751806892: New connection from 192.168.32.106:63803 on port 1883.
1751806892: Client remote5 already connected, closing old connection.
1751806892: New client connected from 192.168.32.106:63803 as remote5 (p2, c1, k10,
u'testuser').
```

```
devel@pi4-37:~/pico_w-mqtt/pi_tcp_tests $ iperf -c remote5
```

```
[ 3] local 192.168.32.1 port 53534 connected with 192.168.32.106 port 5001
[ID] Interval      Transfer  Bandwidth
[ 3] 0.0000-10.0782 sec 17.8 MBytes 14.8 Mbits/sec
```